

# SIMATS ENGINEERING

## ASSESSMENT TOOL 2

COURSE NAME: Database Management Systems  
COURSE CODE: CSA05

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### COURSE OUTCOME COVERED

CO1: Analyze DBMS architecture, different data models, schema types, and design ER/EER diagrams, relational schemas for case-based scenarios by identifying entities and relationships. (BL1, BL2)

Activity Tool : Concept Mapping (Medium)  
Assessment Tool Weightage: 35%

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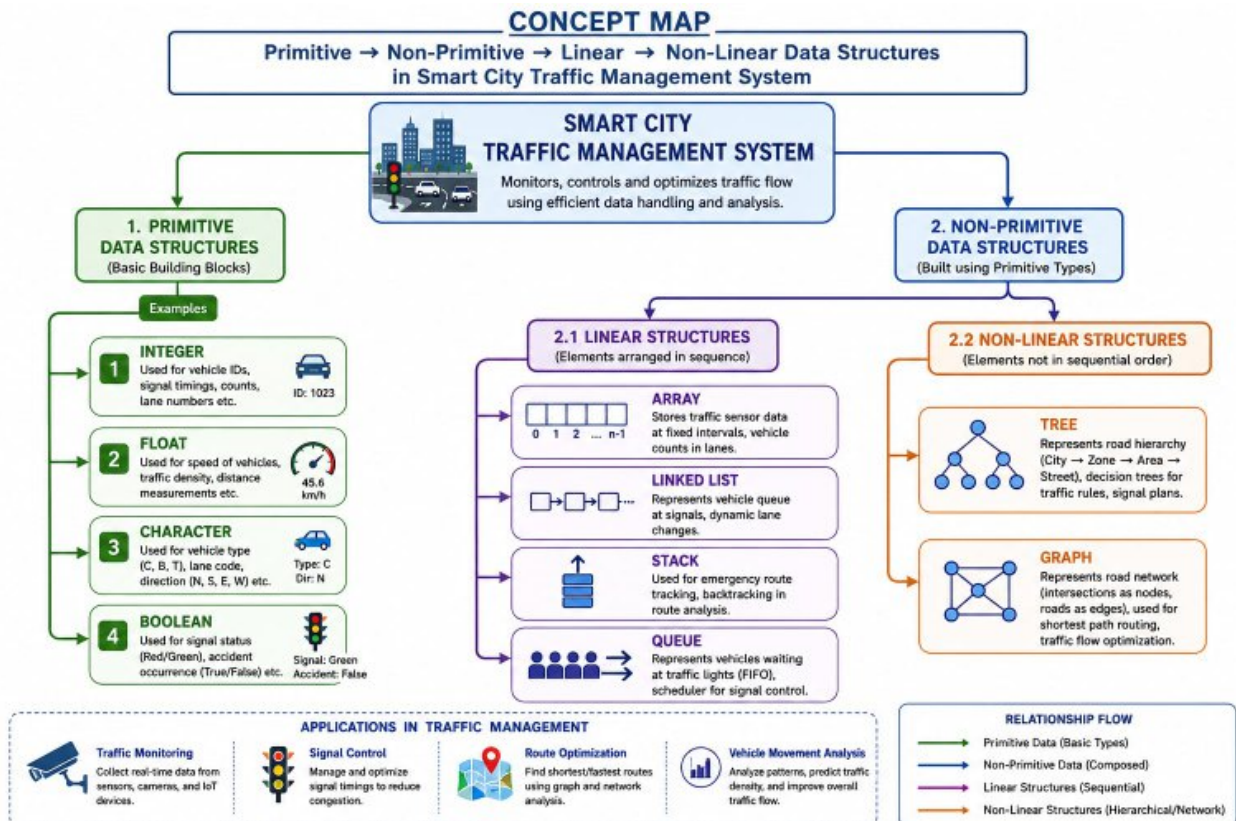
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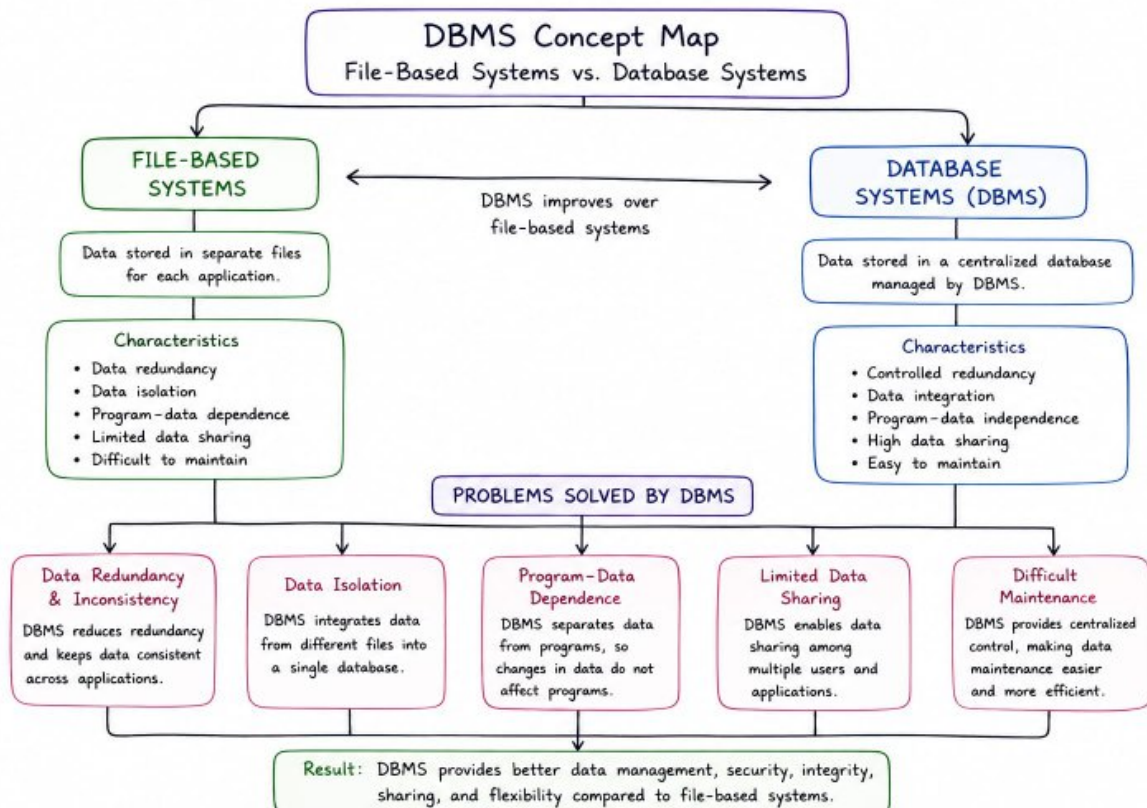
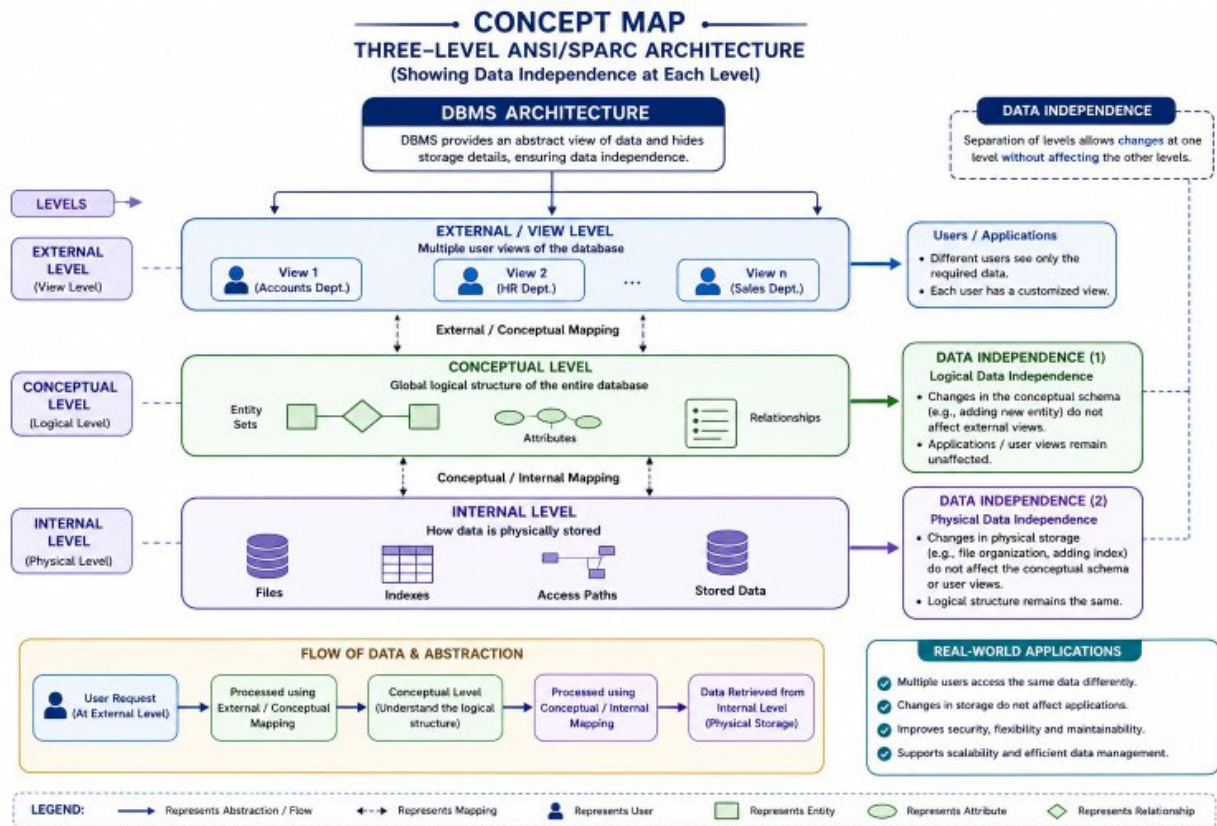
### QUESTIONS

Total Marks: 50

| Q.No | Question                                                                                                                                 | Bloom's Level    | Marks |
|------|------------------------------------------------------------------------------------------------------------------------------------------|------------------|-------|
| 1    | Construct a concept map linking the following terms: DBMS, Schema, Instance, Data Model, ER Diagram, Entity, Attribute, Relationship.    | BL2 – Understand | 5     |
| 2    | Create a concept map for the three-level ANSI/SPARC architecture showing data independence at each level.                                | BL2 – Understand | 5     |
| 3    | Map the relationship between file-based systems and database systems, highlighting the problems solved by DBMS.                          | BL2 – Understand | 5     |
| 4    | Design a concept map for ER diagram components: entity, weak entity, attribute types, relationship types, and participation constraints. | BL2 – Understand | 5     |
| 5    | Construct a concept map comparing Hierarchical, Network, and Relational data models with their key properties.                           | BL2 – Understand | 5     |
| 6    | Develop a concept map for the EER model showing how it extends the basic ER model with generalization, specialization, and aggregation.  | BL2 – Understand | 5     |

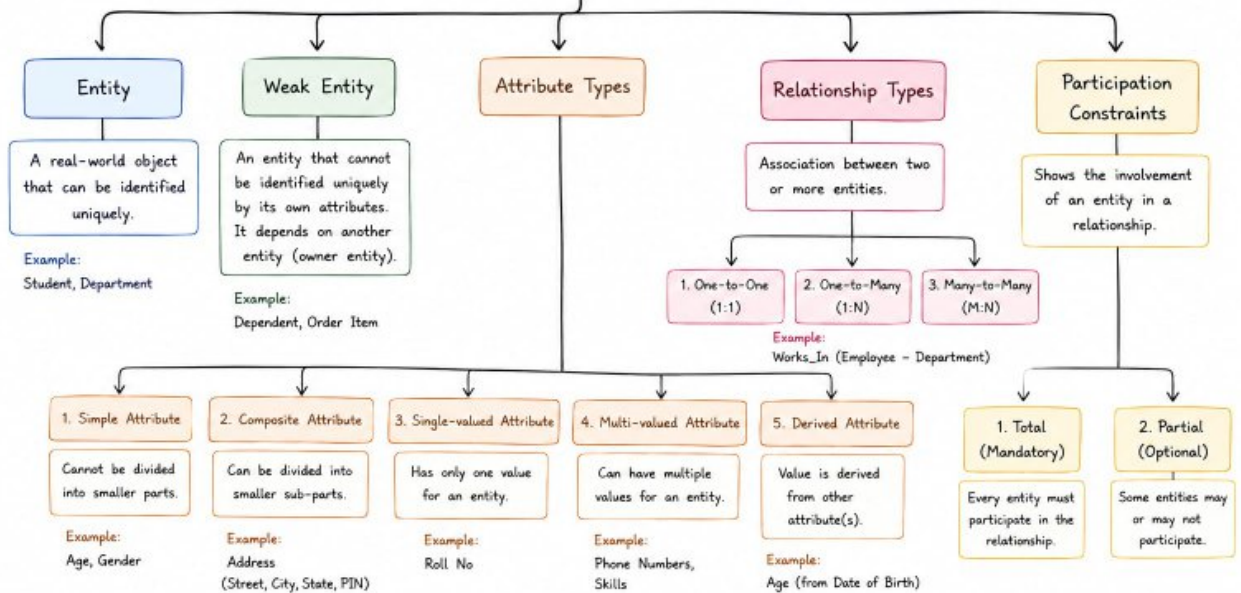
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|----|----------------------------------------------------------------------------------------------------------------------------|------------------|---|
| 7  | Create a visual concept map for the DBMS software architecture, illustrating the flow from user request to data retrieval. | BL2 – Understand | 5 |
| 8  | Map the data models discussed in CO1 to real-world applications and explain the suitability of each model.                 | BL2 – Understand | 5 |
| 9  | Construct a concept map for schemas and instances, showing how they relate to physical and logical data independence.      | BL2 – Understand | 5 |
| 10 | Design a concept map for the roles and responsibilities in a DBMS environment: DBA, End User, Application Developer.       | BL1 – Remember   | 5 |







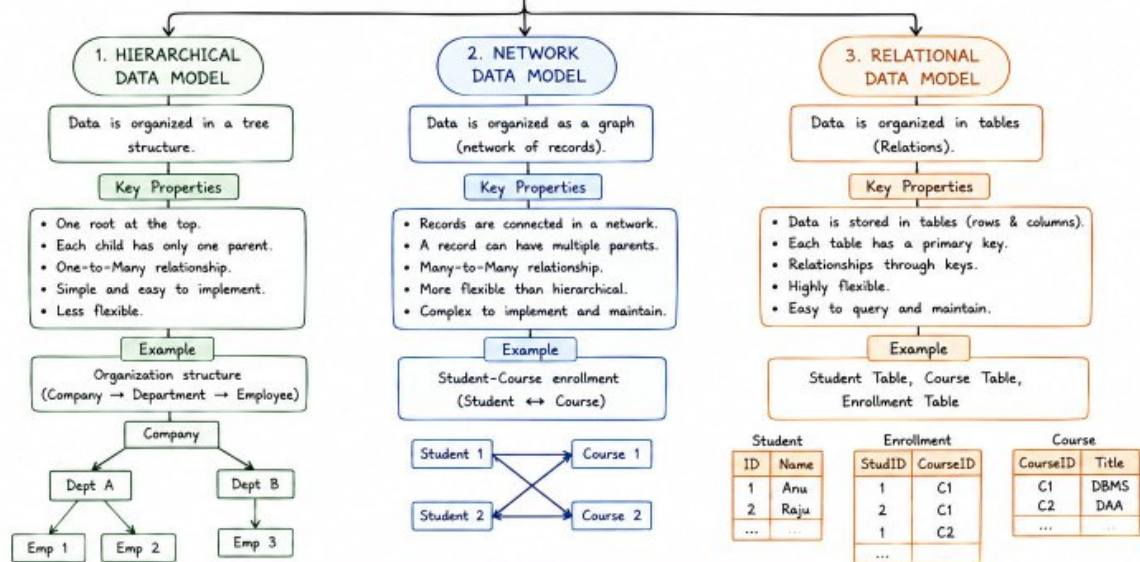
## ER Diagram Components



In ER Diagrams, these components are used together to represent data and the relationships between real-world objects in a clear and structured way.

## DATA MODELS (Comparison)

### Ways to organize data in a database



**Summary :** Hierarchical → Simple tree structure (One-to-Many)  
 Network → Graph structure (Many-to-Many)  
 Relational → Table structure (Relations using keys)

Code of Conduct:

I N.Thejasree (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

N.Thejasree

RUBRICS FOR CALCULATION:

| Criteria               | Weightage | Level 4<br>(Exemplary)                                                                                                                                   | Level 3<br>(Proficient)                           | Level 2<br>(Developing)                                     | Level 1<br>(Beginning)                  |
|------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| Concept Identification | 25%       | Identifies all key concepts from literature accurately and comprehensively                                                                               | Identifies most key concepts with minor omissions | Identifies limited concepts; some are irrelevant or missing | Fails to identify essential concepts    |
| Linkages               | 25%       | Establishes clear, meaningful, and accurate relationships between concepts<br>Concepts are well-structured hierarchically with logical flow and grouping | Shows correct relationships with minor gaps       | Relationships are weak, unclear, or partially incorrect     | No meaningful relationships established |
| Organization           | 20%       | Concepts are well-structured hierarchically with logical flow and grouping                                                                               | Mostly organized with minor structural issues     | Some organization but lacks clear hierarchy                 | No logical organization or structure    |

|                           |     |                                                                                                                                                       |                                               |                                                     |                                         |
|---------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------|
| Integration of Literature | 20% | Integrates multiple studies effectively to show connections and comparisons<br>Concept map is clear, neat, visually structured, and easy to interpret | Shows some integration of literature          | Limited integration; mostly isolated concepts       | No integration of literature evidence   |
| Clarity                   | 10% |                                                                                                                                                       | Generally clear with minor readability issues | Clarity is reduced; difficult to interpret in parts | Poorly presented and hard to understand |

